Maximum Mark: 10 **Quiz-4** **Time: 55 minutes**

1. (points) [2+2]
   (a) Write all the possible L $(=\sum_i l_i)$, S and J values for a configuration containing two equivalent optically active electrons np^2 (treating electron interactions and $\vec{L}.\vec{S}$ as perturbation). Finally list all the possible fine structure terms $(^{2S+1}L_J)$.
   (b) Draw the energy level splitting configuration for the above state ((a)).

2. (points) [2+1] Assuming periodic boundary conditions, derive the expression for number of electrons state within the energy E and E+dE. Using this find the average energy of electrons at T=0 K.

3. (points) [3 (0.5+1+1.5)]
   (a) What is the lifetime of the ground state of He atom?
   (b) Write the expression for the line width for 2p $\rightarrow$ 2s transition in H atom in terms of the life time of the respective state.
   (c) Draw the function and marks the line-width

$$f(\omega) = \frac{(\Gamma^2)/(4\hbar^2)}{(\omega-\omega_0)^2 + (\Gamma^2)/(4\hbar^2)} \tag{1}$$

Useful Results

- $\psi_{100} = \frac{1}{\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2} e^{(-r/a_0)}$;
  $\psi_{200} = \frac{1}{2\sqrt{2\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(1-\frac{r}{2a_0}\right) e^{(-r/2a_0)}$;